

LAYING EGGS

The perch, like many fishes, releases large numbers of eggs to ensure that at least some will eventually produce adult fishes. This is known as broadcast spawning.



LARVA



ADULT

METAMORPHOSIS

When a fish emerges from its egg, its yolk sac is still attached to its body and provides nourishment during the first weeks of its life. The larva of a brown trout (top) takes 3–4 years to mature to the adult form (above).

behavior. However, some fishes conduct complex courtship rituals to increase their chances of attracting a suitable mate. As a prelude to breeding, the male may change color. For example, the male dragonet develops more intense colors, which are displayed to potential mates during elaborate circling movements.

Reproduction

The reproductive behavior of fishes is varied. Although in most cases fertilization takes place outside the female's body and the young emerge from the eggs as larvae, there are many (including all cartilaginous fishes) in which fertilization is internal, and in some of these, the females give birth to live young.

In most fishes (as in many other aquatic animals), fertilization occurs externally. The male releases sperm (milt) over the eggs (roe) as they are released from the female's body. Milt has a thick consistency, which prevents the sperm cells from dispersing too quickly. The eggs of most marine fishes are made buoyant by droplets of oil and float freely as part of the plankton. The eggs of freshwater fishes are typically heavier and have a sticky surface so that they adhere to objects in the water. Some freshwater fishes make a nest for their

eggs, and in some species they guard the eggs (see panel, right). In almost all cases, the eggs hatch to release larvae, which are incompletely formed and gradually develop their skeleton, fins, and some of their organ systems.

External fertilization is a viable way for fishes (and other aquatic animals) to reproduce, partly because the greater density of water makes it a much more suitable medium than air for transferring eggs and sperm. Water also provides the developing eggs with nutrients and dissolved oxygen. However, the chances of an embryo surviving to reach adulthood are relatively low, particularly for marine fishes. To compensate for this, females often produce large numbers of eggs (as many as 5 million in some species).

In all cartilaginous fish, reproduction begins with internal fertilization. In sharks, for example, the inner edges of the pelvic fins are modified into copulatory organs, called claspers, of which one is inserted into the female's cloaca. Sea water flushes the male's sperm along a

groove in this organ and into the female's body.

Where fertilization is internal, the young may be alive when they are released from the female's body. In some

species that give birth to live young, the eggs simply hatch inside the mother's body. In others, including

some species of sharks and rays, there is a connection between the developing embryo and the female, along which nutrients are passed to the embryo.

Live-bearing females invest a relatively large amount of energy in carrying and nourishing their developing young. However, compared with fishes that reproduce

by external fertilization, their young are more developed at birth and have a greater chance of survival. This means that live-bearing fishes can produce and fertilize fewer eggs and yet maintain a stable population.

Compared with other groups of vertebrates, fishes include a relatively large number of hermaphrodites, all of which are bony fishes (see p.487). Some including certain species of carp and loach, reproduce parthenogenetically, the eggs being fertilized without a male. Other species reproduce by a process called gynogenesis. These include the Amazon molly, which is a hybrid of 2 other species and has a population that consists only of females. Reproduction occurs when a female is fertilized by a male of either of the 2 parent species.

Some fishes breed regularly, often once a year. Others, however, reproduce only once in their lifetime, and may die shortly afterward. The timing of reproduction is controlled by external factors (including changes in temperature, light levels, or day length) or by internal cycles (such as changes in hormone levels).

Some fishes have preferred spawning grounds, laying eggs or releasing their young in places where they are most likely to survive. In order to reach these areas, they may have to undertake migrations of thousands of miles.

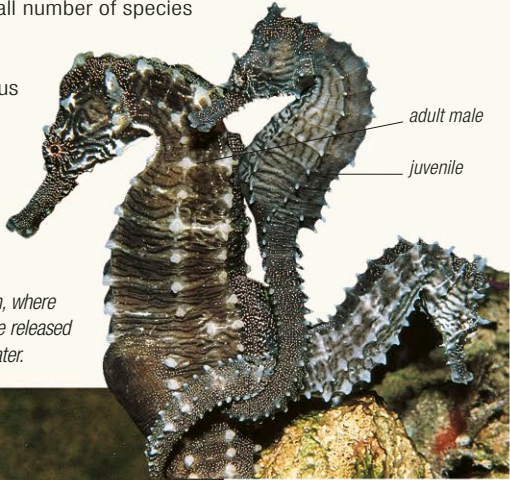
In some species, large numbers of males and females gather in shoals to breed, without any kind of courtship

PARENTAL CARE

Having released their eggs and sperm into water, many fishes have no further contact with their developing young. However, other species actively care for their eggs or young. Besides protecting their eggs by placing them in a nest, adult fishes may care for them by chasing away predators, cleaning them to prevent infection, or fanning them to provide oxygenated water. When the eggs hatch, the parents may use various ways to protect the vulnerable young (see below). A small number of species also feed their young, including the discus fish, which secretes a nutritious fluid through its skin.

SEAHORSES

Male seahorses (right) play an unusual role in caring for their young. The female places her eggs in a pouch in the front of the male's abdomen, where they are fertilized. The young are released when they hatch, 2–6 weeks later.



adult male

juvenile



MOUTH BROODING

Several hundred species of cichlids brood their eggs in cavities in their mouth and throat. When the eggs hatch, the parents also defend the young from predators by allowing them to shelter inside their mouth.



EGGS

Inside its egg, an embryonic fish is nourished by fluids contained in a yolk sac. The egg case is transparent (allowing the eyes of these salmon embryos to be seen from the outside). The time taken for the eggs to hatch often depends on the water temperature.